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Level 5 – Network Operating Systems

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Logbook 6

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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded

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1. INTRODUCTION

1.1 Operating System (OS)

A program that controls the hardware of a computer is called an operating system. It serves as a bridge between computer users and hardware and offers a foundation for application programs. The variety of ways operating systems can carry out these functions is an incredible feature.

The main purpose of mainframe operating systems is to maximize the use of FTP. Operating systems for personal computers (PCs) can run everything from business apps to advanced games.



Figure 1 Operating System

An operating system must be built piece by piece due to its size and complexity. With precisely defined inputs, outputs, and functions, each of these components ought to be a distinct part of the system. We give a broad summary of the main parts of an operating system in this chapter (Silberschatz, et al., 2006).

1.2 Linux

Linux is an operating system. Linux has been around since the mid-1990s. Linux is actually everywhere on our phones and other devices. The Linux Kernel is like the brain of the operating system because it manages how the computer interacts with its hardware and makes sure everything works smoothly and efficiently (Anon., 2024).

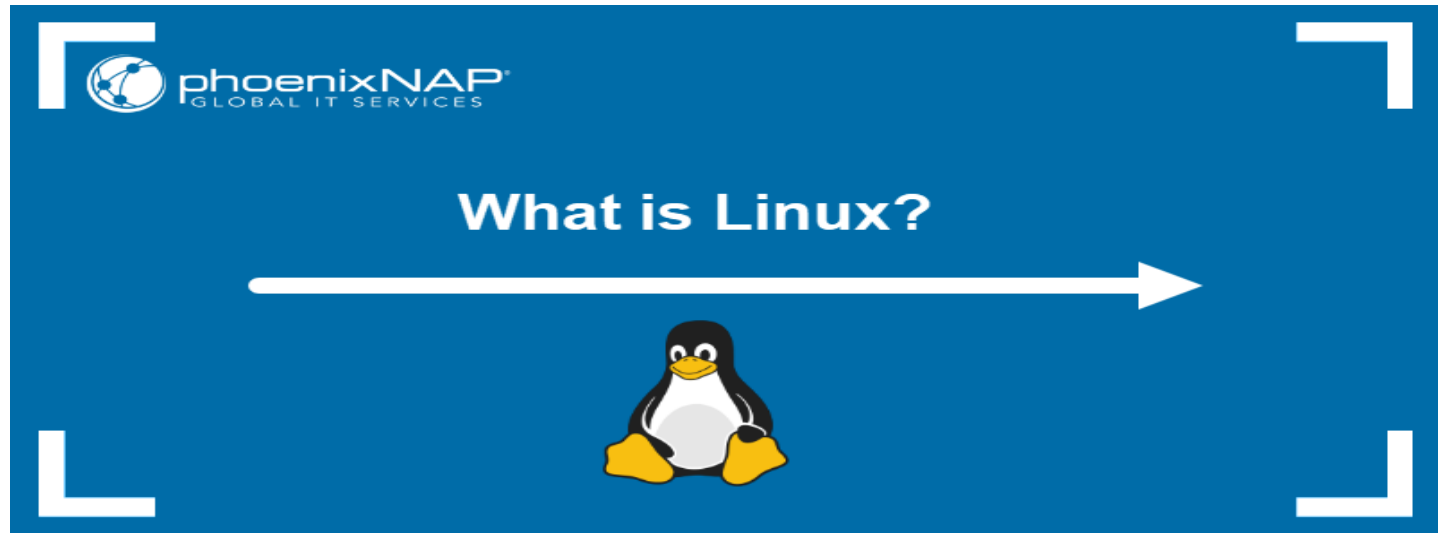


Figure 2 Linux

2. Aims and Objectives

2.1 Aims

The main aims of this logbook 6 is to become a best Debian user by achieve practical experience in managing and configuring the operating system for network-related tasks.

2.2 Objectives

The primary goals and objectives are listed below:

- Learn how to add, remove, and set rules for users.
- Learn how to find and install new programs on Debian.
- Learn how to connect Debian to a network and share the internet.
- To know and understand basic concept to write queries.

3. Required tools and concepts

I used multiple tools to complete my task, which include Ms. Word, snipping tools and Debian. These tools help me a lot. I used MS Word for documentation, the snipping tool for screenshots, and They were so useful and made my work much easier and simpler.

3.1 MS. Word

Microsoft Word is a component of Microsoft Office that facilitates document creation, editing, and formatting. It has tools for text formatting, spell and grammar checkin3.g, and more. In your documents, you can also add charts, tables, and photos. For letters, reports, and resumes, it's common. Word facilitates collaborative document editing by allowing multiple users to track changes. It offers templates to facilitate the creation of documents. Because of its many helpful features and ease of use, users, organizations, and schools frequently use it (Taylor, 24 July 2023).



Figure 3 Ms.Word

3.2a Microsoft Word Features

- **User-friendly interface:** Easy-to-use layout and menu
- **Formatting and templates:** Powerful formatting options and ready-made templates
- **Media integration:** Easily insert images, shapes, charts, and other media elements
- **Collaboration and sharing:** Seamless collaboration, real-time co-authoring, and review/comment features
- **Error checking:** Spelling and grammar checks, Track Changes, and Comment features
- **Page layout and design:** Adjust margins, orientation, headers, and footers
- **Navigation and organization:** Table of Contents, headings, and navigation features



Figure 4 Feature of Ms.Word

3.3 Snipping Tool

Taking screenshots has become one of the necessary components of our daily life. s Snipping Tool makes it simple to take, edit, and share screenshots. Users can enhance screenshots by adding text, highlights, and annotations. I was able to take screenshots of the assignment's testing portion with the aid of this tool. The snipping tool has been incredibly useful for all my work, not just this assignment (Rusen, 24 March 2023).

Snipping Tool



Figure 5- Snipping Tool

3.4 Debian

Debian is a popular and freely available computer operating system (OS) that uses a Unix-like kernel typically Linux along with other program components. Debian is an open-source software developed nearly by 1000 active programmers around the world which is launched in 1993 by Ian Murdock. This operating system that we have created is called Debian GNU/Linux, or simply Debian for short.

Debian, the "Universal Operating System" stands out among the other Linux distributions for its stability and promising features. Debian Project has continued to operate as an all-volunteer organization dedicated to providing free open-source software. The project has its own constitution, social contract and policy documents for organizing the project . In this logbook 6 we used Debian for writing queries. Debian makes it easier and complete faster (Sheldon, 2022).



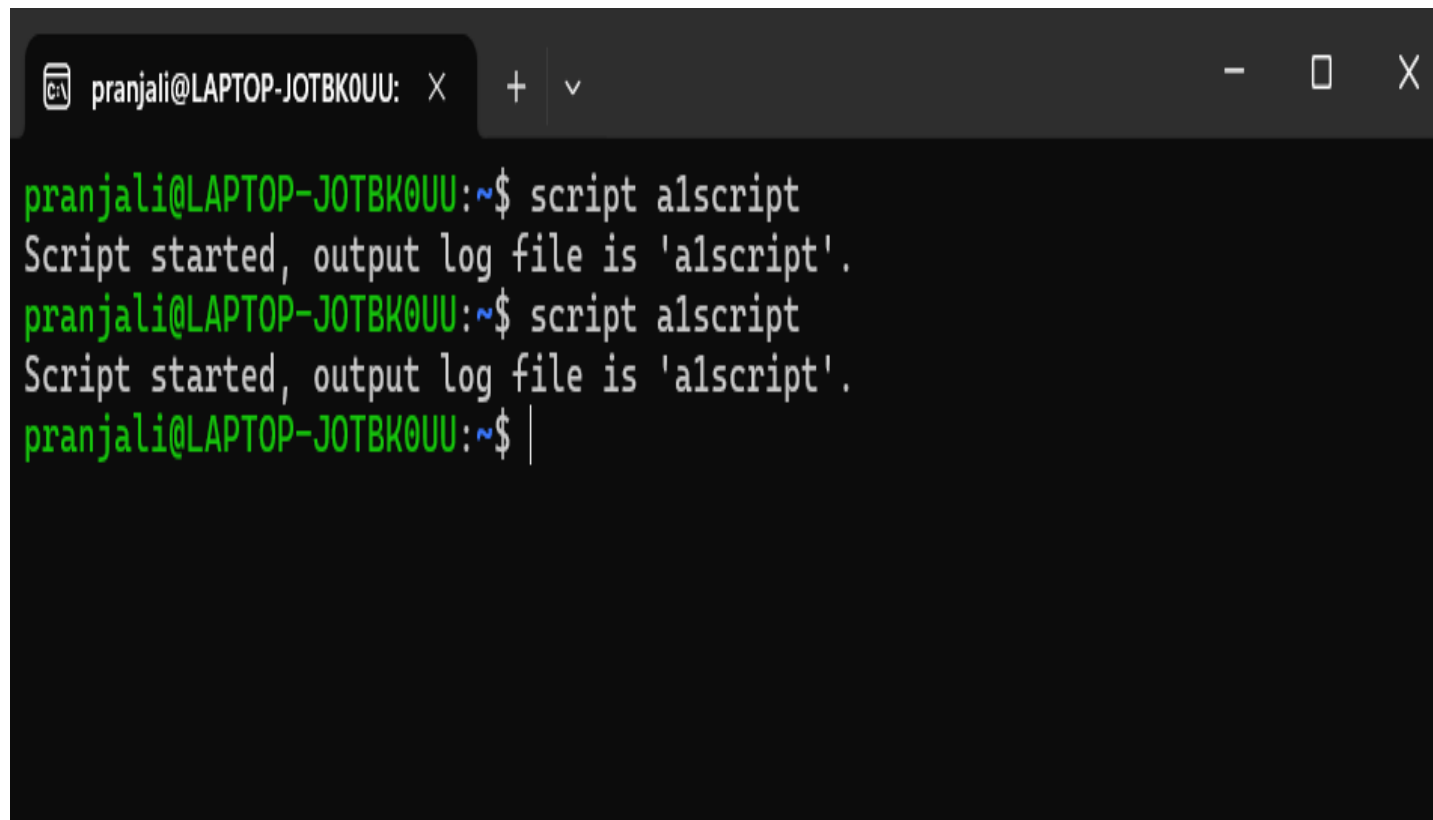
Figure 6 Debian

4. Working on Linux

In this logbook we worked on Linux. The Linux Kernel is like the brain of the operating system because it manages how the computer interacts with its hardware and makes sure everything works smoothly and efficiently. Our teacher gave us several questions, which we wrote down queries in Debian.

5. Questions

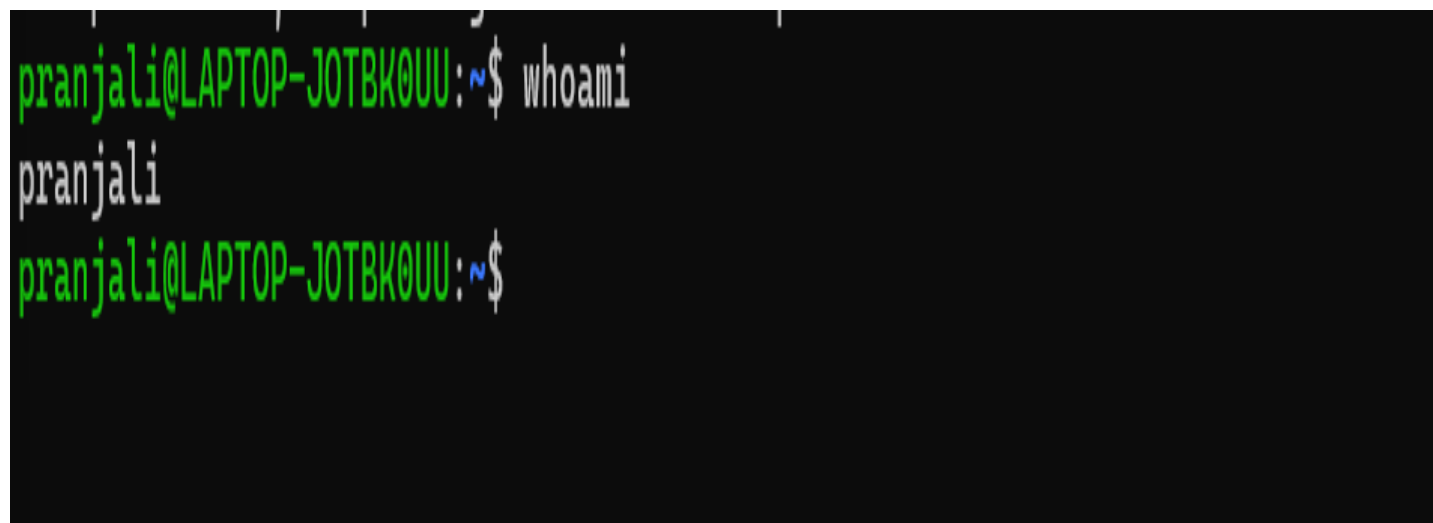
5.1a Type `script a1script` at the prompt. That's the digit one (1) after the letter "a"—this is assignment one, not assignment el. The system will respond with



```
pranjali@LAPTOP-JOTBK0UU: ~$ script alscript
Script started, output log file is 'alscript'.
pranjali@LAPTOP-JOTBK0UU: ~$ script alscript
Script started, output log file is 'alscript'.
pranjali@LAPTOP-JOTBK0UU: ~$ |
```

Figure 7 Command 1

5.1b Type whoami to see your username and Type who to see a list of everyone on the system.

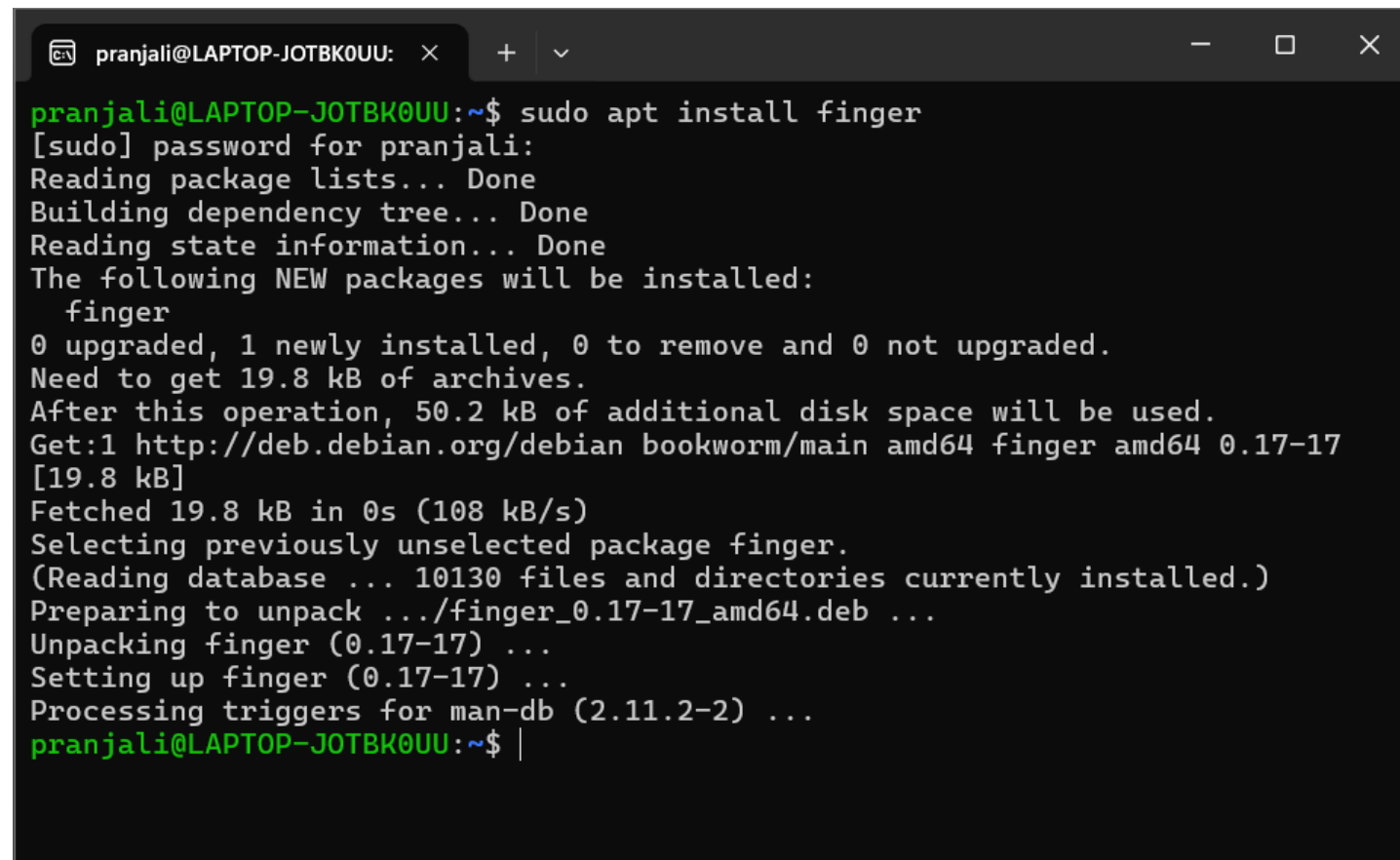


```
pranjali@LAPTOP-JOTBK0UU: ~$ whoami
pranjali
pranjali@LAPTOP-JOTBK0UU: ~$
```

Figure 8 Command 2

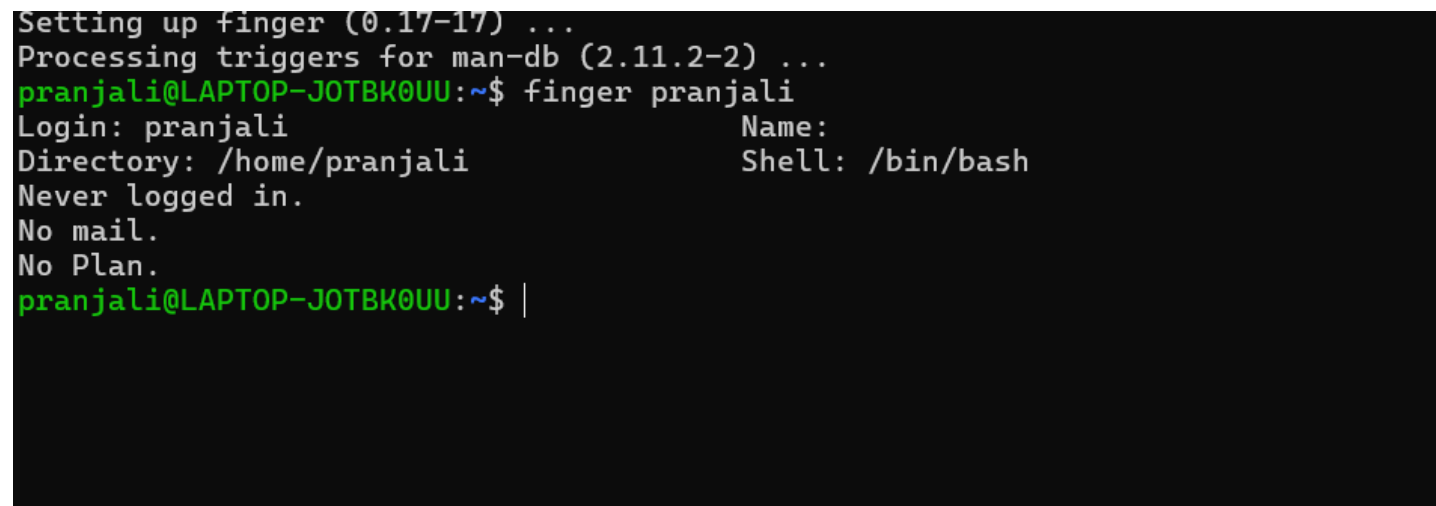
5.1c Type finger linuxnnn, (where linuxnnn is your username) to see more information about

your account.



```
pranjali@LAPTOP-JOTBK0UU: ~$ sudo apt install finger
[sudo] password for pranjali:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  finger
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 19.8 kB of archives.
After this operation, 50.2 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 finger amd64 0.17-17
[19.8 kB]
Fetched 19.8 kB in 0s (108 kB/s)
Selecting previously unselected package finger.
(Reading database ... 10130 files and directories currently installed.)
Preparing to unpack .../finger_0.17-17_amd64.deb ...
Unpacking finger (0.17-17) ...
Setting up finger (0.17-17) ...
Processing triggers for man-db (2.11.2-2) ...
pranjali@LAPTOP-JOTBK0UU: ~$ |
```

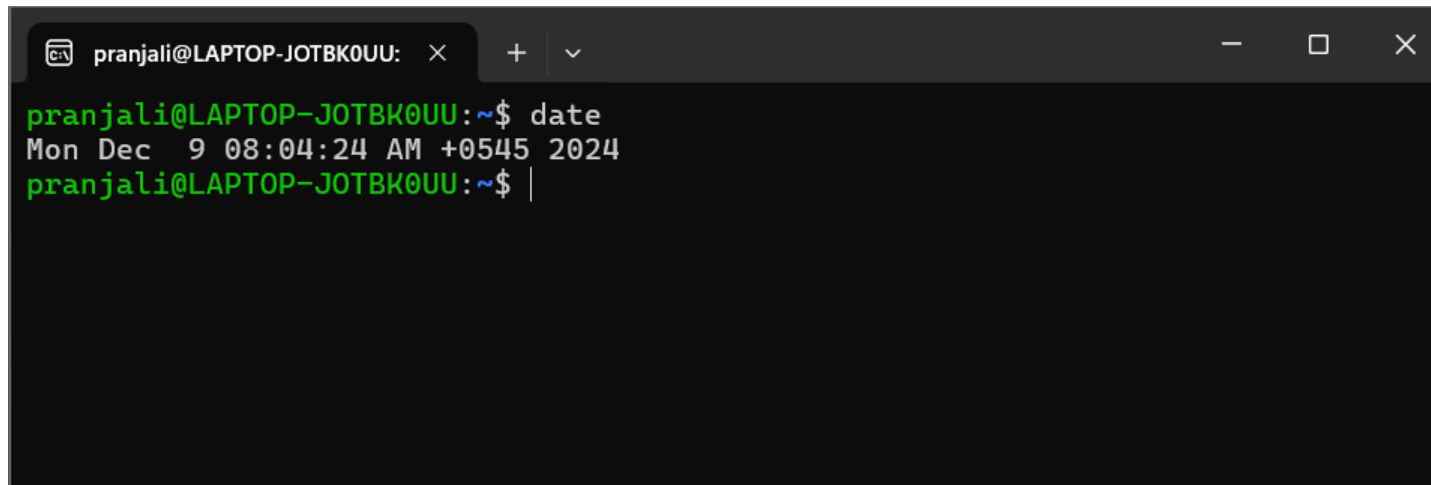
Figure 9 Command 3



```
Setting up finger (0.17-17) ...
Processing triggers for man-db (2.11.2-2) ...
pranjali@LAPTOP-JOTBK0UU: ~$ finger pranjali
Login: pranjali                      Name:
Directory: /home/pranjali           Shell: /bin/bash
Never logged in.
No mail.
No Plan.
pranjali@LAPTOP-JOTBK0UU: ~$ |
```

Figure 10 Command 4

5.1d Type date, to see today's date and the current time.

A terminal window with a dark background. The title bar shows 'pranjali@LAPTOP-JOTBK0UU:'. The prompt is 'pranjali@LAPTOP-JOTBK0UU:~\$'. The command 'date' has been entered, and the output is 'Mon Dec 9 08:04:24 AM +0545 2024'. The prompt is now 'pranjali@LAPTOP-JOTBK0UU:~\$ |' with a cursor.

```
pranjali@LAPTOP-JOTBK0UU:~$ date
Mon Dec 9 08:04:24 AM +0545 2024
pranjali@LAPTOP-JOTBK0UU:~$ |
```

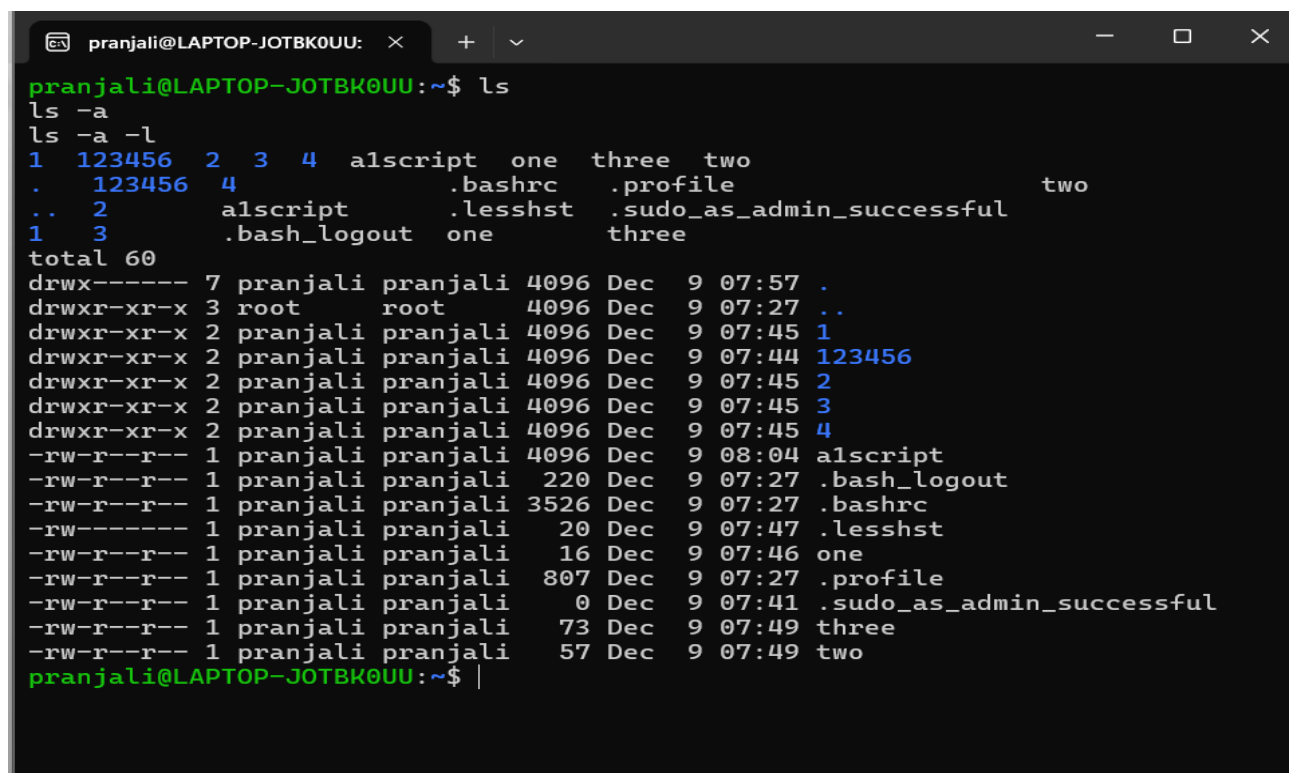
Figure 11 Command 5

5.1e What files do you have? Type these three commands. Each one produces a different output.

ls,

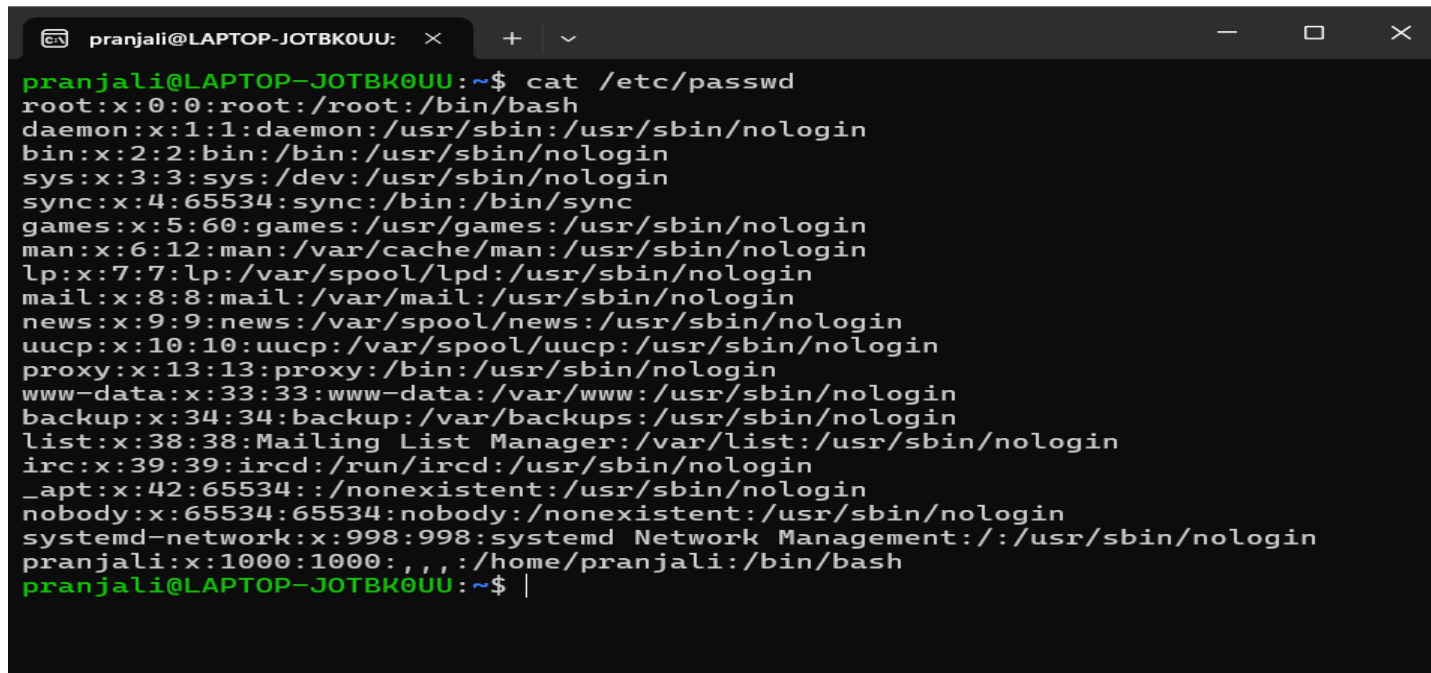
ls -a,

ls -a -l

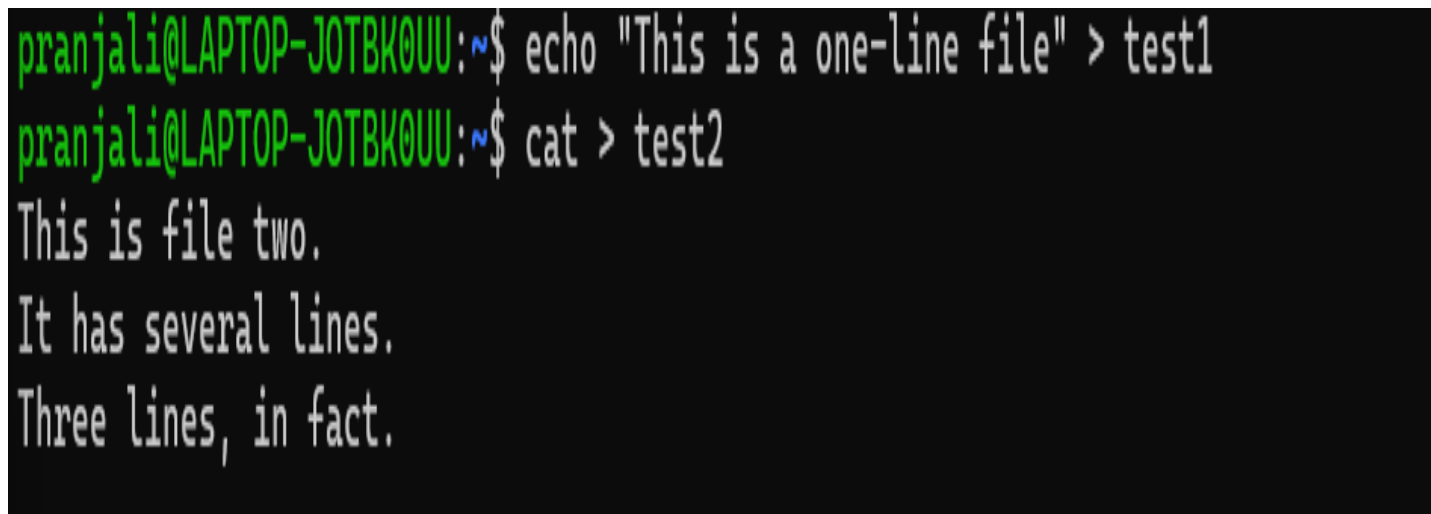
A terminal window with a dark background. The title bar shows 'pranjali@LAPTOP-JOTBK0UU:'. The prompt is 'pranjali@LAPTOP-JOTBK0UU:~\$'. The command 'ls' has been entered, and the output is a list of files and directories. The prompt is now 'pranjali@LAPTOP-JOTBK0UU:~\$ |' with a cursor.

```
pranjali@LAPTOP-JOTBK0UU:~$ ls
ls -a
ls -a -l
1 123456 2 3 4 alscript one three two
. 123456 4 .bashrc .profile two
.. 2 alscript .lessht .sudo_as_admin_successful
1 3 .bash_logout one three
total 60
drwx----- 7 pranjali pranjali 4096 Dec 9 07:57 .
drwxr-xr-x 3 root root 4096 Dec 9 07:27 ..
drwxr-xr-x 2 pranjali pranjali 4096 Dec 9 07:45 1
drwxr-xr-x 2 pranjali pranjali 4096 Dec 9 07:44 123456
drwxr-xr-x 2 pranjali pranjali 4096 Dec 9 07:45 2
drwxr-xr-x 2 pranjali pranjali 4096 Dec 9 07:45 3
drwxr-xr-x 2 pranjali pranjali 4096 Dec 9 07:45 4
-rw-r--r-- 1 pranjali pranjali 4096 Dec 9 08:04 alscript
-rw-r--r-- 1 pranjali pranjali 220 Dec 9 07:27 .bash_logout
-rw-r--r-- 1 pranjali pranjali 3526 Dec 9 07:27 .bashrc
-rw-r--r-- 1 pranjali pranjali 20 Dec 9 07:47 .lessht
-rw-r--r-- 1 pranjali pranjali 16 Dec 9 07:46 one
-rw-r--r-- 1 pranjali pranjali 807 Dec 9 07:27 .profile
-rw-r--r-- 1 pranjali pranjali 0 Dec 9 07:41 .sudo_as_admin_successful
-rw-r--r-- 1 pranjali pranjali 73 Dec 9 07:49 three
-rw-r--r-- 1 pranjali pranjali 57 Dec 9 07:49 two
pranjali@LAPTOP-JOTBK0UU:~$ |
```

Figure 12 Command 6

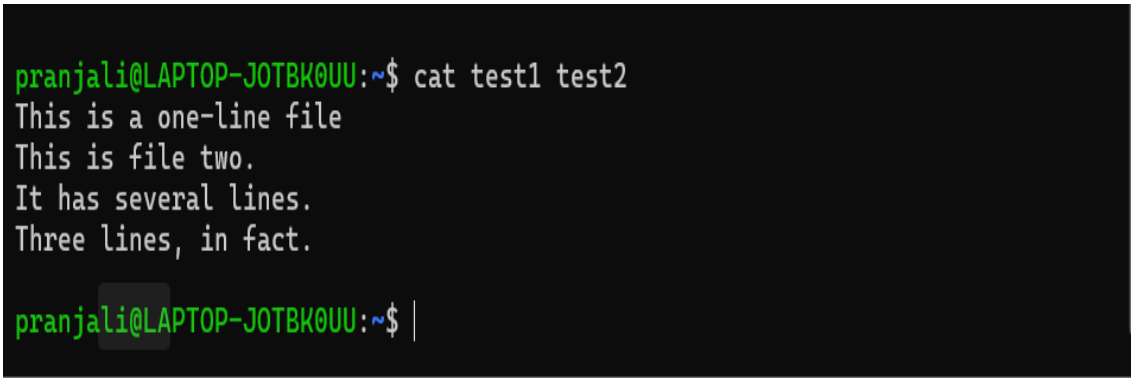
5.1f Type below commands. cat /etc/passwd

```
pranjali@LAPTOP-JOTBK0UU: ~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
pranjali:x:1000:1000:,,,:/home/pranjali:/bin/bash
pranjali@LAPTOP-JOTBK0UU: ~$ |
```

*Figure 13 Command 7***5.1g Create a file named test1 by typing this: echo "This is a one-line file" > test1**

```
pranjali@LAPTOP-JOTBK0UU: ~$ echo "This is a one-line file" > test1
pranjali@LAPTOP-JOTBK0UU: ~$ cat > test2
This is file two.
It has several lines.
Three lines, in fact.
```

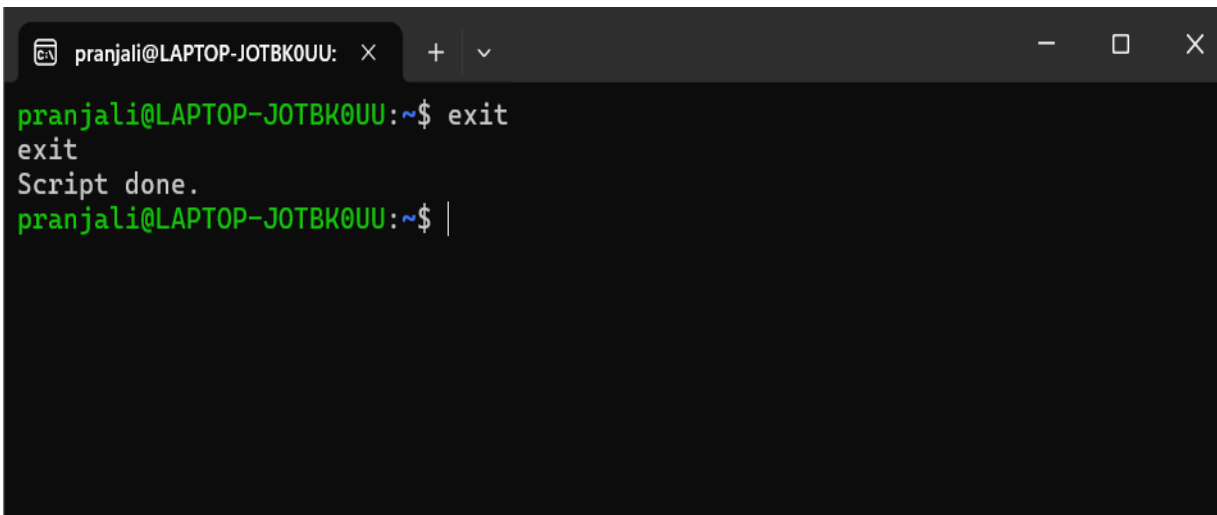
*Figure 14 Command 8***5.1h Create another file by typing the following, where ^D means CTRL-D. cat > test2, This is file two. It has several lines. Three lines, in fact. ^D and Combine test1 and test2 file**

A terminal window with a black background and green text. The prompt is 'pranjali@LAPTOP-JOTBK0UU:~\$'. The command 'cat test1 test2' has been entered. The output consists of four lines: 'This is a one-line file', 'This is file two.', 'It has several lines.', and 'Three lines, in fact.'. The prompt is now 'pranjali@LAPTOP-JOTBK0UU:~\$' with a cursor at the end.

```
pranjali@LAPTOP-JOTBK0UU:~$ cat test1 test2
This is a one-line file
This is file two.
It has several lines.
Three lines, in fact.
pranjali@LAPTOP-JOTBK0UU:~$ |
```

Figure 15 Command 9

5.1i exit

A terminal window with a black background and green text. The prompt is 'pranjali@LAPTOP-JOTBK0UU:~\$'. The command 'exit' has been entered. The output is 'exit' followed by 'Script done.'. The prompt is now 'pranjali@LAPTOP-JOTBK0UU:~\$' with a cursor at the end.

```
pranjali@LAPTOP-JOTBK0UU:~$ exit
exit
Script done.
pranjali@LAPTOP-JOTBK0UU:~$ |
```

Figure 16 Command 10

6.Conclusion

In this logbook, we get to know a lot about Debian. We worked in this using commands, setting up users, and connecting to networks. We also learn how to install programs and keep Debian learning how to work on my Debian system.

The logbook 6 demonstrated the useful application of fundamental networking ideas and offered invaluable practical experience. This logbook has not only deepened my understanding of Debian but has also improved my technical writing skills. I've learned to document my actions, observations, and troubleshooting steps in a clear and concise manner, which will be invaluable in my future endeavor.

7.References

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